

Contact

allenlazer9@gmail.com

www.linkedin.com/in/geoffrey-lazer-09b0901b2 (LinkedIn)

Top Skills

Data Engineering

Long Short-term Memory (LSTM)

Wildfire

Certifications

Salz21 AI Hackathon Hackathon Certificate

From Data to Insights with Google Cloud

R Programming for Machine Learning and Data Science

Object Detection and Recognition using Deep Learning and OpenCV

HTML, Wordpress and CSS for Internet Marketers

Geoffrey Lazer

ML Engineer | MSc Computing Science at University of Alberta | AI/ML Researcher | XR & Unity Developer
Edmonton, Alberta, Canada

Summary

I'm a Computing Science graduate student at the University of Alberta (Multimedia + AI/ML), passionate about building immersive and intelligent systems that connect AI, XR, and data visualization.

Previously, I've worked on 3D volumetric rendering of DICOM datasets in Unity/C#, developed reinforcement learning models for glucose prediction, and designed interactive dashboards that visualize complex datasets. My technical foundation includes data structures, OOP, and system design, combined with hands-on experience in Unity XR development, AI/ML model design, and full-stack web engineering.

Achievements: Winner – Smart India Hackathon 2022 | Finalist – Niral Thiruvizha 2024

Core Skills: Unity XR • C# • Python • PyTorch • TensorFlow • React • Flask • PostgreSQL • GCP • OpenCV • Vertex AI • System Design

Experience

Firesafe AI

Machine Learning Engineer

October 2025 - May 2026 (8 months)

Edmonton, AB

Luxsonic Technologies Inc.

Software Developer Internship

May 2025 - August 2025 (4 months)

Saskatoon, Saskatchewan, Canada

- Developed 3D volumetric visualization of DICOM data in Unity for real-time slicing and transfer functions [\[PDF\]](#).
- Built a gesture-based VR keyboard (OpenXR) enabling efficient text input in immersive environments.

- Integrated offline voice-to-text (Vosk) for real-time VR transcription, creating a multimodal input system.
- Explored mesh extraction pipelines (Dicom2Mesh, C++/VTK) to optimize medical imaging for lightweight rendering.

University of Alberta

Graduate Teaching Assistant

January 2025 - April 2025 (4 months)

Edmonton

CMPUT 402:

- Supported a senior-level software engineering course focused on software quality, DevOps, and automation.
- Guided students in building CI/CD pipelines with GitHub Actions, integrating JUnit testing, static analysis (SonarQube), mutation testing (Pitest), and code coverage tools to ensure code reliability.
- Mentored teams on DevOps best practices including automated deployment, feature toggles, and A/B testing.
- Reviewed projects for technical debt, security, and performance considerations, reinforcing industry-grade quality standards.
- Collaborated closely with students and instructors, developing strong communication and mentorship skills that align with cross-functional teamwork in professional environments.

MedWatch Technologies, Inc.

AI/ML Researcher

October 2024 - December 2024 (3 months)

Edmonton, Alberta, Canada

- Built Reinforcement Learning models to improve blood glucose prediction accuracy.
- Developed two DQN models that tuned device configurations, improving MARD over an LSTM baseline by reducing 7 percent error variance.
- Designed architecture diagrams; deployed solutions on Google Cloud Platform (GCP).
- Tech stack: Python, TensorFlow, PyTorch, RLlib, GCP.

TechCzar

Head of Design

July 2023 - September 2023 (3 months)

Chennai

- Lead the design team and assigned roles to each team member
- Designed the bi-annual magazine for the department and was responsible for managing proofreading and editing

Spacescan Ltd

Intern

September 2022 - February 2023 (6 months)

Canada

- Worked closely with the Front End Team to enhance UI experience and design
- Solved the website's bugs to ensure the right information is displayed for each NFT

LICET PATTARAI

Director of Business Development

November 2021 - December 2022 (1 year 2 months)

Chennai, Tamil Nadu, India

- Collaborated with various Entrepreneurs and Business Leaders to avail funding, industrial projects and knowledge transfer sessions
- Managed the budget for the club to finance club events

Education

University of Alberta

Master's degree, Computing Science, Multimedia · (September 2024 - July 2026)

Loyola-ICAM College of Engineering and Technology

Bachelor of Engineering - BE, Computer Science and Engineering · (July 2020 - June 2024)

St. Xavier's School, Jaipur

High School Diploma, PCM-IP · (March 2008 - March 2020)

Gefion Gymnasium - STX

High School Diploma, Student Exchange Program · (April 2018 - June 2018)